

Does Showing a Language Model the Downside Change the Reward Code It Writes?

A preregistered controlled experiment on the feedback channel of automated reward design, in risk sensitive portfolio reinforcement learning

Tamer | Academic supervisor: Dr Ramin Okhrati | Industrial supervisor: Stefan Wagner

1. The question

A reinforcement learning agent optimises whatever its reward says. In risk sensitive portfolio control, writing that reward is the hard part, because an objective specified in the mean can produce a policy that behaves for years and then loses badly once.

Eureka showed that a language model can write reward code and improve it from feedback, and that the feedback carries weight: remove the reflection step and about 29% of the performance goes with it. Across that whole line of work the feedback is a single number. It reports how the last attempt scored and says nothing about the shape of what happened.

Shape is what matters here. On our panel the daily cross section has excess kurtosis of 15.25, empirical conditional shortfall runs from 0.84 to 1.66 times the Gaussian benchmark as the tail deepens, and on the worst days about a fifth of the names fall together. One number at one level cannot carry a tail that reverses across quantiles. Six can. Whether a model actually uses them is another matter.

Research question. *When a language model writes the reward code for a trading agent, does showing it the tail of the realised outcome distribution, rather than a single score, change the code it writes, and does any change reach the agent's realised risk?*



2. What we build on, and what we change

Two papers set the frame. **Eureka** is the loop we adapt: the model writes a reward, the agent trains on it, a score comes back, the model revises. The **Decision Language Model** is the closest structural precedent, because its designer iterates on a distribution rather than a scalar, although that distribution is a population spread in public health with no risk axis and no fixed agent underneath. Behind both sits the optimal reward problem: the reward you hand an agent is a design variable, not the objective you actually care about.

We change one thing, the feedback. The five language model conditions share one fixed Soft Actor Critic agent, one budget of thirty candidates, and two prompts that are byte identical and verified free of tail, shortfall, drawdown, and quantile vocabulary. Only the block returned after each attempt differs, and the control block is a coordinate projection of the treatment block. Nothing else moves. A complicated manipulation would be a fragile one, so the complexity sits in the checks rather than in the design.

Selection is blind to the tail on purpose. Candidates are picked on a validation Deflated Sharpe ratio with the risk aversion weight at zero, computed from realised returns, so it cannot be reward hacked and favours no arm. The tail is fed on train, selection runs on validation, and the test window opens once. Three estimators, three partitions, no arm grading itself. Rewards see anonymised arrays with integer indices, never tickers or dates. One caveat stays in the open: the fed tail comes from the trained policy's own returns, so it is endogenous, and the comparison is between two coupled loops of reward, policy, and measurement.

3. The setup

Agent	Stable Baselines3 Soft Actor Critic, twin critics, identical across arms, PopArt normalisation uniform, replay capped at 50,000
Budget	400,000 environment steps per candidate, the measured knee of the learning curve rather than convergence
Universe	top 30 US large caps by point in time market capitalisation plus cash, long only on a softmax simplex
State	60 day return lookback, 20 day volatility and its 20 over 60 ratio, lagged VIX, prior weights, all lagged
Costs	10 bps on turnover, swept from 0 to 50, turnover being half the L1 distance to drifted weights
Data	Refinitiv, 963 securities over 5,406 sessions to June 2026, survivorship free, point in time, delisting to cash
Splits	train 2005 to 2016, validate 2017 to 2019, sealed test 2020 to 2026, 60 session purge at each boundary
Loop	six generations of five candidates per arm, reflecting on the best so far, all prompts and rewards archived for replay

The purge is the greater of a 21 session embargo and the feature lookback, so the February to March 2020 crash falls outside the test leg. Authored code is untrusted: a static allowlist screens it and it runs once in a killable child process.

4. Why it should work, and why it might not

In principle the vector can only help. The scalar is a projection of it, so by Blackwell an optimal designer given the vector does weakly better under any bounded loss and any prior, and by Kusuoka a profile of shortfalls across levels spans the coherent risk class, which a Sharpe number does not.

That is an envelope over optimal users, not a forecast about this one. Three things can keep a real system away from it.

Selection cannot supply the benefit, since the selector is blind. The model has to condition on the numbers, which is empirical rather than assumed. And a mean critic cannot be relied on to turn a tail aware reward into tail protective behaviour, because static shortfall optimality is time inconsistent, needs state augmentation, and has an optimal policy that is generally non Markovian. The null is over determined. That is why the null is what we registered, and why the interesting question is which link fails.

5. The conditions, in priority order

	Arm	Feedback after each attempt	Why it is in the design
Primary	distributional scalar	score plus the six statistic tail block score alone	the treatment the comparison that answers the question
Controls	placebo scalar_cvar5	score plus an inert block matched in length score plus one downside number	rules out length and token count rules out any single downside number
Reference	placebo_shuffled random_search bayes_opt	the treatment structure, values deranged no model, sampling over a reward family no model, Bayesian search over a template	format against content how much outcome is due to reward design whether open ended code beats tuning

The first four arms carry the hypothesis and are the last to be cut. The search baselines matter differently: without them nobody can say how much of a difference comes from reward design rather than the agent and the market, and no published work in this line shows an authored reward beating matched compute black box search.

6. What is tested, and how it is decided

The headline is H2: does tail feedback beat scalar feedback at matched compute? It is decided by two co primary tests, one on Sharpe and one on shortfall at 5%, each requiring the treatment to beat all three comparators one sided at 5%. Outcomes are per seed, aggregated by interquartile mean, contrasted by a paired bootstrap over shared seeds. Demanding all three legs is itself the multiplicity correction. Three secondaries sit around it, reported without gating anything: authored rewards against four hand written baselines, iteration against a single shot of thirty, and the model against the two search baselines.

Equivalence matters as much as rejection here. The margin is 0.05 in deflated Sharpe units. Pilot seed noise is 0.369 on the paired Sharpe difference, so at the thirty seed floor the minimum detectable effect is about 0.120, well above the margin, and a non rejection there reads inconclusive rather than equivalent. The ladder climbs through seven rungs to 568 seeds, with 403 the registered target at 95% assurance. The stop is exogenous, set by cluster throughput against the deadline, never by looking at results. Overfitting is handled by combinatorially symmetric cross validation, and a factor ladder to six factors plus betting against beta and quality minus junk answers the low volatility objection.

7. Locating the break

This is the part that makes a null worth reading. Responsiveness asks whether movement in the fed numbers moves a structural feature of the authored program, with a positive control of about twenty overtly directive stimuli where the statistic must fire, so a flat result cannot be blamed on a dead instrument. Transmission is a mediation decomposition, read descriptively since the mediator is endogenous. Specificity compares abstract syntax tree shapes with identifiers stripped, so a program that merely names a variable after a risk statistic is separable from one that computes differently. If the chain breaks at the first link, the leading suspect is legibility. Frontier models compare close small floats at roughly 50 to 70% accuracy, and the fed values sit in that regime: the marginal error on a fed shortfall level is about 0.0033, while paired differences between candidates run from 0.0001 to 0.0008. Some of what the designer is asked to read is near its own noise floor. So the same content is re rendered as integer basis points and rank labels. A separated positive differential confirms the bottleneck, a null refutes it, and a probe that adds one instruction sentence separates cannot read from will not use.

8. Eleven models, two tiers

The verdict is two tier by design. Opus 4.8 is the confirmatory instrument, because ladder depth power against the equivalence margin is affordable for exactly one model. Ten further legs run the same five conditions at the thirty seed floor, where any single leg is inconclusive by construction, so they are read jointly rather than one at a time: the sign pattern across legs, a joint per seed permutation test, a pooled bounded interval, the two capability pair differences in differences, and a capability gradient with three signatures named in advance. Generalisation is answered by all eleven, and the mechanism conclusion rests on all eleven plus the reading survey, not on Opus alone. Making all eleven confirmatory would mean roughly sixty six corrected tests, twenty times the compute, and an inference target nobody can define, since there is no population of language models to sample from.

The legs are DeepSeek V4 Pro, GLM 5.2, Qwen 3.6 27B and Qwen 3.5 9B as a dense open capability pair, Nemotron 3 Super, Haiku 4.5 as the closed pair against Opus, Sonnet 5, GPT 5.6 Luna, Gemini 3.5 Flash, and Kimi K3. Five are open weight and pinned to a repository commit for the exact weights, with served quantisation disclosed and decoding fixed, which is what lets someone re run the experiment rather than only the analysis. A reading survey then asks whether models can read these numbers at all, across 35 models and roughly ten labs at no GPU cost. One extension is specified but not activated and still undecided: it would measure each model's just noticeable difference for numeric comparison and report the share of realised fed deltas below it, closing the mechanism story with a number rather than an argument.

Two limits are worth stating plainly. The claim is comparative between feedback channels, never that the system beats the market, and the fed signal is six left tail scalars spanning the coherent risk class rather than the return distribution.